

Uptake and Biotransformation of Sevoflurane in Humans: A Comparative Study of Sevoflurane with Halothane, Enflurane, and Isoflurane

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Study Objective: To compare the volatile anesthetic sevoflurane with halothane, enflurane, and isoflurane on the uptake and biotransformation in humans.

Design: Prospective pharmacokinetic study of sevoflurane administration in human subjects.

Setting: Inpatient surgery clinic at a university medical center.

Patients: Thirty-two Japanese patients, free of systemic diseases, undergoing minor elective surgery with endotracheal general anesthesia.

Interventions: The patients were assigned randomly to one of four groups: halothane, enflurane, isoflurane, or sevoflurane. One of the four volatile anesthetics being investigated [equivalent to 1.1 minimum alveolar concentration (MAC): halothane, 0.85%; enflurane, 1.85%; isoflurane, 1.27%; and sevoflurane, 1.88%; in inspired concentrations throughout the first hour of anesthesia] was administered for 60 minutes.

Measurements and Main Results: In all patients, serum and urinary fluoride concentrations were measured. The concentrations of all gases were measured separately with a mass spectrometer. The cumulative uptake of each anesthetic agent during a certain period was calculated as an integration of the uptake rate per minute. The results for one-hour inhalation of sevoflurane (1.1 MAC) showed an uptake (corrected for body surface area and MAC) of 490 ml/m²/MAC and estimated degradation rate of 3.3%. For purposes of comparison, similar studies of halothane (uptake, 653 ml/m²/MAC; degradation rate 15.7%), enflurane (1150 ml/m²/MAC; 1.3%), and isoflurane (439 ml/m²/MAC; 0.6%) were also conducted. Sevoflurane had a peak serum inorganic fluoride concentration of 19.3 μmol/L, and no abnormality in hepatic or renal functions was observed in any of the subjects during the two weeks postoperatively.

Conclusions: Accurate determinations of uptake and degradation rate for sevo-

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flurane and three other volatile anesthetics in Japanese patients were obtained. These findings have established that, despite its relatively large MAC (1.71%), sevoflurane has a small uptake due to its low solubility. However, the degradation rate was shown to be as high as 3.3%, resulting in a higher serum fluoride concentration than seen after administration of isoflurane, halothane, and (possibly) enflurane.

Keywords: Anesthesia; sevoflurane; halothane; enflurane; isoflurane; biotransformation; pharmacokinetics; mass spectrometry.

Introduction

Sevoflurane, a new volatile anesthetic, has not yet been approved for clinical use and is currently undergoing clinical trials in the United States and Japan. The most notable characteristic of sevoflurane is its low blood-gas partition coefficient, which is lower than that of the other volatile anesthetics currently in clinical use. Although its MAC is relatively large (about the same as that of enflurane), the uptake of sevoflurane is expected to be small due to its low solubility. There has been only one published report on the metabolism of sevoflurane.¹ However, the measurements of the uptake of sevoflurane in that study may not be accurate, since the subjects were anesthetized by mask, which does not exclude the possibility of leaks. In addition, the uptake and degradation rate were measured in healthy subjects, and these measurements would differ from those in surgical patients given the anesthetic under clinical conditions. There are two reasons for this difference: first, surgical patients experience an alteration of blood flow to organs (*e.g.*, cardiac output, pulmonary blood flow, hepatic blood flow, and renal blood flow) as a result of the invasive procedure of surgery; second, healthy volunteers soon return to their normal daily activities after exposure to the anesthetic in a controlled experiment, whereas surgical patients are necessarily confined to bed rest following surgery. Furthermore, racial differences affecting uptake must be taken into account.

To clarify these questions, the current study was undertaken to obtain accurate measurements of the clinical uptake and degradation rates of sevoflurane in surgical patients anesthetized and intubated. To substantiate the accuracy of the measurement system used in this study, similar comparative studies were conducted of three other volatile anesthetics currently in clinical use (halothane, enflurane, and isoflurane), of which the uptakes and degradation rates are known.

Materials and Methods

Thirty-two ASA I patients who were scheduled to undergo minor elective surgery served as subjects. After informed consent was obtained, the patients were assigned randomly to one of four groups: halothane, enflurane, isoflurane, or sevoflurane. Each group consisted of an equal number of males and females, except for the enflurane group (three males, five females). The subjects ranged in age from 19 to 55 years (35.4 ± 11.3 years). The mean height was 159.5 ± 8.7 cm, the mean weight was 56.4 ± 8.7 kg, and the mean body surface area was 1.59 ± 0.15 m². All were found to be free of systemic disease on the basis of routine preoperative laboratory tests.

An intravenous (IV) infusion of lactated Ringer's solution was maintained throughout anesthesia. Anesthesia was induced with thiopental sodium (5 mg/kg), and tracheal intubation followed paralysis with succinylcholine chloride (2 mg/kg). Pancuronium bromide was given after tracheal intubation, and patients' lungs were ventilated mechanically with 50% nitrous oxide, 30% oxygen, and 20% nitrogen for 20 to 30 minutes to maintain the arterial carbon dioxide tension (PaCO₂) at 30 to 35 mmHg and to minimize the second gas effect. One of the four volatile anesthetics being investigated (equivalent to 1.1 MAC: halothane, 0.85%; enflurane, 1.85%; isoflurane, 1.27%; and sevoflurane, 1.88%; in inspired concentrations throughout the first hour of anesthesia) was then administered for 1 hour. Additional pancuronium was given as needed. After 1 hour, a combination of diazepam and fentanyl was administered for the next hour, during which the volatile anesthetic was eliminated. Vital signs—blood pressure (BP), heart rate (HR), electrocardiogram (EKG), and rectal temperature—were monitored throughout the anesthesia.

An arterial catheter was inserted into a radial artery for continuous BP monitoring and measurement of blood gases and serum fluoride. Blood gases were measured with a Model ABL-II blood gas analyzer (Radiometer, Copenhagen, Denmark). Arterial blood was sampled just before administration of the volatile anesthetic and 15, 30, 45, 60, 75, 90, 105, 120, and 240 minutes and 24, 48, 72, and 96 hours after administration. Daily collection of urine was begun the day before surgery and continued for 4 days postoperatively. The urine was tested for inorganic fluoride, specific gravity, pH, creatinine, electrolytes, and nitrogen. Serum and urine specimens were kept at -20°C for later analysis of fluoride. Arterial blood samples were taken during the morning on postoperative days 1, 7, and 14. Urinalysis also was performed on postoperative days 1 through 4, 7, and 14.

The fluoride ion concentration was determined by a fluoride ion specific electrode² (Model 96-09, Orion Research, Inc., Cambridge, MA) in conjunction with an ion analyzer (Model EA940, Orion Research, Inc., Cambridge, MA). Total urinary fluoride content was measured by a modified oxygen-flask combustion method³ (Model FHO 1-A, Hamada-Rika, Osaka, Japan). The level of fluoride ion concentration from organic compounds was calculated as the difference between the total urinary fluoride and the inorganic fluoride ion concentrations.

The cumulative amount of excessive fluoride excretion primarily represents the metabolic process of the inhaled anesthetic. It was assumed that 50% of fluoride metabolite formed was absorbed by bone.⁴ Thus, the degradation rate of the inhaled anesthetic was estimated from the following equations:

$$\text{Organic fluorine (OF)} = \text{total urinary fluoride (TF)} \\ - \text{inorganic fluoride (IF)}$$

$$\text{Excessive amount} = \text{amount excreted per day} \\ - \text{baseline value}$$

$$\text{Degradation rate} = \frac{\text{cumulative (excessive OF} \\ + 2 \times \text{excessive IF)}}{\text{number of fluorine atoms} \\ \times \text{cumulative uptake}}$$

The concentrations of end-tidal, mixed inspired, and mixed expired gases were monitored continuously by the previously described system.⁵ The concentrations of the volatile anesthetic agent, carbon dioxide, nitrous oxide, nitrogen, and oxygen were measured separately with a mass spectrometer (Medical Gas Analyzer 1100, Perkin Elmer, Pomona, CA), precalibrated with the use of the volatile anesthetic agents of known concentration predetermined by a gas chromatograph (Model GC-9A, Shimadzu, Kyoto, Japan). The expiratory volume per minute was measured continuously with a recently developed linearized electric Wright respirometer (BOC Medishield, Essex, England).

On the assumption that the fractional concentration of the end-tidal anesthetic is equal to that in the alveolar gas (F_A), the ratio of F_A to that in the inspired gas (F_I) can be calculated. The uptake rate of the anesthetics ($\dot{V}_{ANE}$), in ml/min, was calculated according to the following equations:

$$\dot{V}_{ANE} = \dot{V}_I \times F_{IANE} - \dot{V}_E \times F_{EANE}$$

$$\dot{V}_I = \dot{V}_E \times \frac{F_{EN_2}}{F_{IN_2}}$$

where $\dot{V}_I$ = the ventilation volume of the inspiratory gas per minute; $\dot{V}_E$ = the ventilation volume of the

expiratory gas per minute; F_I = the fraction of the inspiratory gas, and F_E = the fraction of expiratory gas.

To get $\dot{V}_I$, the anesthetic was mixed with an inert gas such as nitrogen from room air, as suggested by Sullivan.⁶ Because oxygen uptake is not equal to carbon dioxide elimination, $\dot{V}_I$ is not equal to $\dot{V}_E$. Therefore, the respiratory quotient is not 1.0. The cumulative uptake of each anesthetic agent during a certain period was calculated as an integration of the uptake rate per minute.

The mean values and standard deviations for each group were calculated. Logarithmic transformation was used to stabilize the variance of the means over time. Statistical intragroup and intergroup comparisons were made using the Bartlett test. These data were then analyzed by one-way analysis of variance (ANOVA). Student's *t*-test, corrected for multiple analyses (Bonferroni), was used when differences were found with ANOVA. A *p* value less than 0.05 was considered significant.

Results

The patients in each group were comparable in age, height, weight, blood loss, fluid intake, urinary output, and duration of surgery and anesthesia. No patient received any blood transfusion. Arterial blood gases in all patients were within normal limits, and the vital signs (BP, HR, EKG, and rectal temperature) were stable in all patients throughout anesthesia and thereafter. No abnormal values were observed in clinical laboratory tests during the 2 weeks following surgery. Blood urea nitrogen, creatinine, and transaminase concentrations also were within the normal range.

Table 1 shows the results of the F_A/F_I ratios after 1 hour of inhalation (Figure 1), the respective cumulative uptakes (BTPD: Body Temperature Pressure Dry) after 1 hour of inhalation (Figure 2), the amounts eliminated in the expiratory air during the hour following 1 hour of inhalation, and the total uptakes of the anesthetic agents which were corrected for body surface area and MAC (Figure 3).

The average serum inorganic fluoride concentration preoperatively was about 1 $\mu\text{mol/L}$ in all four groups: halothane, $1.2 \pm 1.0 \mu\text{mol/L}$; enflurane, $1.2 \pm 0.6 \mu\text{mol/L}$; isoflurane, $1.3 \pm 1.1 \mu\text{mol/L}$; and sevoflurane $1.3 \pm 0.8 \mu\text{mol/L}$. There was no significant difference in preoperative fluoride among groups. The inhalation of halothane and isoflurane for 1 hour did not result in any significant change in the serum inorganic fluoride concentration. Follow-

Table 1. Total Uptake and Other Results of This Study

	Total Uptake for 1 Hour (ml)	Normalized Uptake (ml/m ² /MAC)	Elimination via the Lung for 1 Hour (%)	F _A /F _I Ratio after 1 Hour (%)
Halothane (n = 8)	797.1 ± 270.4	653.0 ± 184.0	18.1 ± 4.2	66.9 ± 5.7
Enflurane (n = 8)	1,344.8 ± 397.8	1,150.0 ± 221.6	23.5 ± 7.1	71.3 ± 4.8
Isoflurane (n = 8)	552.0 ± 98.2	439.3 ± 103.6	25.6 ± 10.4	72.2 ± 3.9
Sevoflurane (n = 8)	702.9 ± 130.1	490.0 ± 122.4	40.2 ± 12.8	84.5 ± 3.0

Data are means ± SD.

MAC = minimum alveolar concentration; F_A/F_I = alveolar gas / inspired gas concentration ratio.

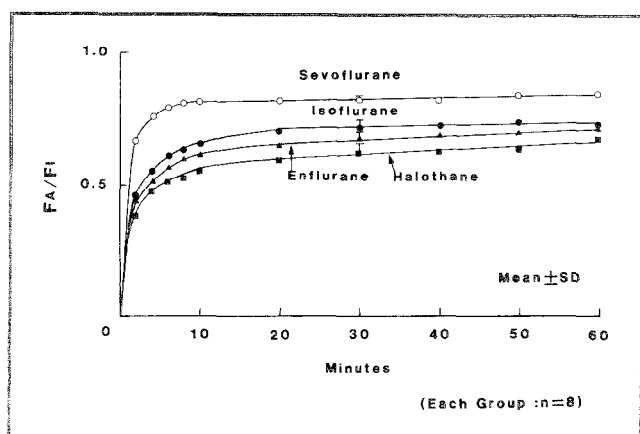


Figure 1. The alveolar/inspiratory concentration ratios (F_A/F_I) in different anesthetics were plotted as a function of time.

ing the inhalation of enflurane, the concentration reached its maximum value, $12.5 \pm 3.3 \mu\text{mol/L}$, 2 hours after the cessation of enflurane inhalation, decreasing gradually to the baseline value on day 3. Following sevoflurane inhalation, the concentration peaked at $19.3 \pm 3.8 \mu\text{mol/L}$ 30 minutes after the cessation of inhalation, decreasing to the baseline level on day 2 (Figure 4). The total amounts of fluoride eliminated, the ratios of the fluoride eliminated to the total uptake, and the ratios of the amounts of organic fluoride to that of inorganic fluoride eliminated into the urine for 5 days postoperatively of sevoflurane inhalation were smaller than those for halothane inhalation but larger than those for enflurane and isoflurane inhalation (Table 2). No complications or adverse effects on renal and hepatic functions were observed with the use of sevoflurane.

Discussion

The uptake of sevoflurane has generated substantial interest among anesthesiologists because sevoflurane

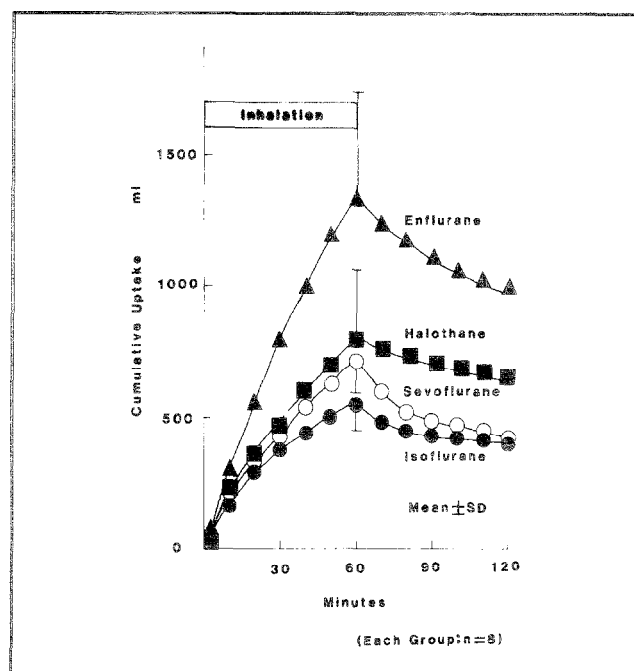


Figure 2. The cumulative uptake and the elimination of the different anesthetic agents during and after inhalation. The cumulative uptakes in 60 minutes are the total uptake of the agents, and those uptakes in 120 minutes are residual amounts not eliminated via the lung. Total uptake of sevoflurane differed significantly from that of enflurane ($p < 0.05$) but showed no significant difference from that of isoflurane ($p < 0.05$).

is less soluble than some other volatile anesthetics and yet has a relatively large MAC. Compared with enflurane,⁷ which has a smaller MAC, sevoflurane naturally has a smaller uptake due to its low solubility. Sevoflurane also has a smaller uptake than halothane,^{8,9} which has a smaller MAC. Furthermore, sevoflurane has a lower degradation rate than halothane, affording even less possibility of organ toxicity and thus making sevoflurane the more useful of the two anesthetics. Isoflurane,^{10,11} which has a smaller MAC

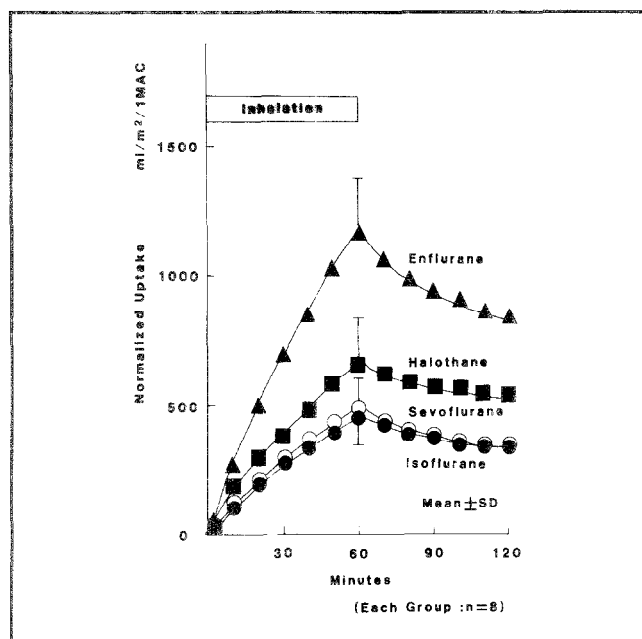


Figure 3. The uptake and the elimination of anesthetic agents during and after inhalation were normalized by body surface area and minimum alveolar concentration (MAC).

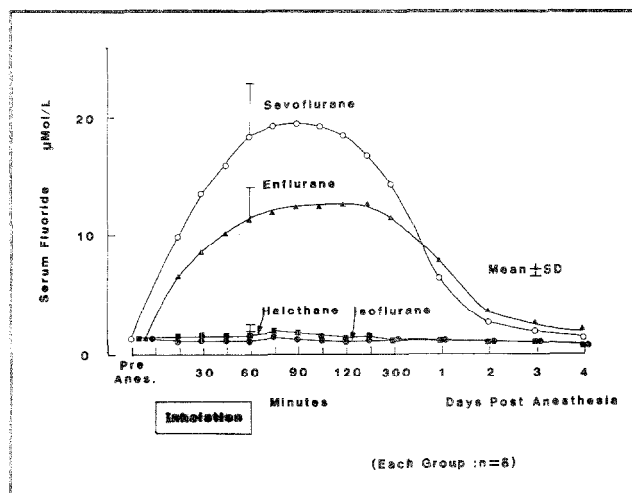


Figure 4. The serum inorganic fluoride concentrations were plotted against time. Note that those concentrations in sevoflurane and enflurane rose as high as 20 $\mu\text{mol/L}$, then decreased to normal within 2 days postoperatively. Statistical difference ($p < 0.05$) was observed from 15 to 120 minutes of the commencement of inhalation.

Table 2. Metabolites as a Percent of Total Uptake

	Total Amount of Fluoride in Urine for 5 days (mmol)	Elimination in Urine as a Percent of Total Uptake (%)	Estimated Degradation Rate as a Percent of Total Uptake (%)	Ratio of Organic to Inorganic Fluoride in Urine
Halothane (n = 8)	14.6 $\pm$ 9.2	15.7 $\pm$ 10.0	15.7 $\pm$ 10.0	11.9
Enflurane (n = 8)	2.6 $\pm$ 1.1	1.0 $\pm$ 0.5	1.3 $\pm$ 0.7	3.3
Isoflurane (n = 8)	0.6 $\pm$ 0.4	0.5 $\pm$ 0.3	0.6 $\pm$ 0.4	4.9
Sevoflurane (n = 8)	5.9 $\pm$ 3.1	2.9 $\pm$ 1.5	3.3 $\pm$ 1.7	6.1

Data are means $\pm$ SD.

than sevoflurane, has about the same level of uptake but a lower degradation rate. These findings show that sevoflurane is a volatile anesthetic that is absorbed in small amounts when administered in clinical concentrations.

In the calculation of the uptake of a volatile anesthetic in previous studies, only the amount expired was measured. In contrast, the method used in this study enabled the measurement of the amount inspired as well. Both expired air and inspired air were directed into a mixing chamber, and nitrogen was used as a tracer.⁶ Thus, more accurate determinations of uptake were possible. This method also enabled continuous sampling without interrupting the anesthesia. In previous studies, spot sampling was conducted on gases collected in a Douglas bag, a method

that has two disadvantages that may adversely affect the accuracy of the data: (1) the possibility of gas adsorption onto the Douglas bag and (2) the impossibility of continuous sampling, which means that the data must be obtained by interpolation.

The F_A/F_I ratios for the other volatile anesthetic agents tested were inversely proportional to their respective blood-gas partition coefficients, in agreement with previously published results.¹²⁻¹⁴ Thus, the data obtained for sevoflurane can be presumed to be accurate: sevoflurane has the highest F_A/F_I ratio of the agents tested.

Holaday and Smith¹ did not report a definite degradation rate for sevoflurane. The accurate determination of the uptake of sevoflurane in the current study made it possible to calculate the degradation

rate of sevoflurane for the first time and to compare it with the degradation rate of other volatile anesthetics. Taking into account the amount absorbed by bone, the authors found the degradation rate of sevoflurane to be $3.3\% \pm 1.7\%$.

The maximum serum inorganic fluoride ion concentration of sevoflurane was $19.3 \pm 3.8 \mu\text{mol/L}$, well below the margin of safety of $50 \mu\text{mol/L}$.¹³ No hepatic or renal damage was observed in any of the subjects. However, the results might be different in a group of patients at greater risk—for example, those given a prolonged anesthetic, obese patients, enzyme induction, similar to that described in the literature for enflurane.

The urinary fluoride O/I ratio of sevoflurane was 6.1 in the current study. Since one molecule of sevoflurane has seven fluorine atoms, the authors believe that one of the fluorine atoms undergoes dehalogenation to an inorganic fluoride ion during the degradation of sevoflurane.

In summary, accurate determinations of the uptake and degradation rate for sevoflurane and three other volatile anesthetics in Japanese patients were obtained. The findings have established that despite its relatively large MAC (1.71%), sevoflurane has a small uptake due to its low solubility. However, the degradation rate was shown to be as high as 3.3%, resulting in a higher serum fluoride concentration than was seen after administration of isoflurane, halothane, and (possibly) enflurane.

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